# ClipSAT : IB Diploma Mathematics: Analysis & Approaches / Applications

## Unit 8.01: Topic 8: IB SL Paper 1 & Paper 2 Extended-Response: Synthesis

Name: _____

**Q1.** The graph of the function fx = -x^2 + 4x - 3 is shown below. What are the coordinates of the y-intercept?

[ ] A: 0, -3
[ ] B: 0, 3
[ ] C: -1, 0
[ ] D: 3, 0

**Q2.** Consider the function fx = 2x + 3 . What is the value of f4 ?

[ ] A: 7
[ ] B: 11
[ ] C: 10
[ ] D: 5

**Q3.** Consider the function fx = 2x - 3 . Find the value of f3 .

[ ] A: 3
[ ] B: 6
[ ] C: 9
[ ] D: 0

**Q4.** Consider the function fx = -x^2 + 6x - 5 . Determine the coordinates of the vertex of the graph of y = fx .

[ ] A: 3, 4
[ ] B: -3, 4
[ ] C: 3, -4
[ ] D: 4, 3

**Q5.** Consider the function fx = 2x - 1 . Find the value of f3 .

[ ] A: 5
[ ] B: 6
[ ] C: 7
[ ] D: 4

**Q6.** Consider the function fx = x^2 - 4x + 3 . The graph of y = fx is shown. Determine the value of f0 .

[ ] A: 1
[ ] B: -1
[ ] C: 3
[ ] D: -3

**Q7.** Consider the function fx = -2x + 6 . The graph of y = fx is shown below. Determine the value of the y -intercept.

[ ] A: 6
[ ] B: -3
[ ] C: 3
[ ] D: -6

**Q8.** Consider the function fx = x^2 - 4x + 3. The graph of y = fx is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

[ ] A: 1, 0
[ ] B: 3, 0
[ ] C: 0, 3
[ ] D: 0, 1